In [298…

```python
import pandas as pd
import seaborn as sb
import matplotlib.pyplot as plt
```

# Load Dataset

In [301…

```python
df=pd.read_csv(r"D:\Downloads\loan_approval_dataset.csv")
```

In [303…

```python
df
```

Out[303…

| | loan_id | no_of_dependents | education | self_employed | income_annum | loan_amount | loan_term | cibil_score | residential_assets_value | commercial_assets_value | luxury_assets_value | bank_asset_value | loan_stat |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **0** | 1 | 2 | Graduate | No | 9600000 | 29900000 | 12 | 778 | 2400000 | 17600000 | 22700000 | 8000000 | Approv |
| **1** | 2 | 0 | Not Graduate | Yes | 4100000 | 12200000 | 8 | 417 | 2700000 | 2200000 | 8800000 | 3300000 | Reject |
| **2** | 3 | 3 | Graduate | No | 9100000 | 29700000 | 20 | 506 | 7100000 | 4500000 | 33300000 | 12800000 | Reject |
| **3** | 4 | 3 | Graduate | No | 8200000 | 30700000 | 8 | 467 | 18200000 | 3300000 | 23300000 | 7900000 | Reject |
| **4** | 5 | 5 | Not Graduate | Yes | 9800000 | 24200000 | 20 | 382 | 12400000 | 8200000 | 29400000 | 5000000 | Reject |
| **...** | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | |
| **4264** | 4265 | 5 | Graduate | Yes | 1000000 | 2300000 | 12 | 317 | 2800000 | 500000 | 3300000 | 800000 | Reject |
| **4265** | 4266 | 0 | Not Graduate | Yes | 3300000 | 11300000 | 20 | 559 | 4200000 | 2900000 | 11000000 | 1900000 | Approv |
| **4266** | 4267 | 2 | Not Graduate | No | 6500000 | 23900000 | 18 | 457 | 1200000 | 12400000 | 18100000 | 7300000 | Reject |
| **4267** | 4268 | 1 | Not Graduate | No | 4100000 | 12800000 | 8 | 780 | 8200000 | 700000 | 14100000 | 5800000 | Approv |
| **4268** | 4269 | 1 | Graduate | No | 9200000 | 29700000 | 10 | 607 | 17800000 | 11800000 | 35700000 | 12000000 | Approv |

4269 rows × 13 columns

In [305…

```python
df.isna().sum()
```

Out[305…

```
loan_id                     0
 no_of_dependents           0
 education                  0
 self_employed              0
 income_annum               0
 loan_amount                0
 loan_term                  0
 cibil_score                0
 residential_assets_value   0
 commercial_assets_value    0
 luxury_assets_value        0
 bank_asset_value           0
 loan_status                0
dtype: int64
```

In [306…

```python
df[' loan_status']
```

Out[306…

```
0        Approved
1        Rejected
2        Rejected
3        Rejected
4        Rejected
           ...
4264     Rejected
4265     Approved
4266     Rejected
4267     Approved
4268     Approved
Name:  loan_status, Length: 4269, dtype: object
```

In [309…

```python
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4269 entries, 0 to 4268
Data columns (total 13 columns):
 #   Column                     Non-Null Count  Dtype
---  ------                     --------------  -----
 0   loan_id                    4269 non-null   int64
 1    no_of_dependents          4269 non-null   int64
 2    education                 4269 non-null   object
 3    self_employed             4269 non-null   object
 4    income_annum              4269 non-null   int64
 5    loan_amount               4269 non-null   int64
 6    loan_term                 4269 non-null   int64
 7    cibil_score               4269 non-null   int64
 8    residential_assets_value  4269 non-null   int64
 9    commercial_assets_value   4269 non-null   int64
 10   luxury_assets_value       4269 non-null   int64
 11   bank_asset_value          4269 non-null   int64
 12   loan_status               4269 non-null   object
dtypes: int64(10), object(3)
memory usage: 433.7+ KB
```

In [311…

```python
df.columns=df.columns.str.strip()
```

In [313…

```python
df
```

Out[313…

| | loan_id | no_of_dependents | education | self_employed | income_annum | loan_amount | loan_term | cibil_score | residential_assets_value | commercial_assets_value | luxury_assets_value | bank_asset_value | loan_stat |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **0** | 1 | 2 | Graduate | No | 9600000 | 29900000 | 12 | 778 | 2400000 | 17600000 | 22700000 | 8000000 | Approv |
| **1** | 2 | 0 | Not Graduate | Yes | 4100000 | 12200000 | 8 | 417 | 2700000 | 2200000 | 8800000 | 3300000 | Reject |
| **2** | 3 | 3 | Graduate | No | 9100000 | 29700000 | 20 | 506 | 7100000 | 4500000 | 33300000 | 12800000 | Reject |
| **3** | 4 | 3 | Graduate | No | 8200000 | 30700000 | 8 | 467 | 18200000 | 3300000 | 23300000 | 7900000 | Reject |
| **4** | 5 | 5 | Not Graduate | Yes | 9800000 | 24200000 | 20 | 382 | 12400000 | 8200000 | 29400000 | 5000000 | Reject |
| **...** | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | |
| **4264** | 4265 | 5 | Graduate | Yes | 1000000 | 2300000 | 12 | 317 | 2800000 | 500000 | 3300000 | 800000 | Reject |
| **4265** | 4266 | 0 | Not Graduate | Yes | 3300000 | 11300000 | 20 | 559 | 4200000 | 2900000 | 11000000 | 1900000 | Approv |
| **4266** | 4267 | 2 | Not Graduate | No | 6500000 | 23900000 | 18 | 457 | 1200000 | 12400000 | 18100000 | 7300000 | Reject |
| **4267** | 4268 | 1 | Not Graduate | No | 4100000 | 12800000 | 8 | 780 | 8200000 | 700000 | 14100000 | 5800000 | Approv |
| **4268** | 4269 | 1 | Graduate | No | 9200000 | 29700000 | 10 | 607 | 17800000 | 11800000 | 35700000 | 12000000 | Approv |

4269 rows × 13 columns

In [315…

```python
df['loan_status']
```

Out[315…

```
0        Approved
1        Rejected
2        Rejected
3        Rejected
4        Rejected
           ...
4264     Rejected
4265     Approved
4266     Rejected
4267     Approved
4268     Approved
Name: loan_status, Length: 4269, dtype: object
```

In [317…

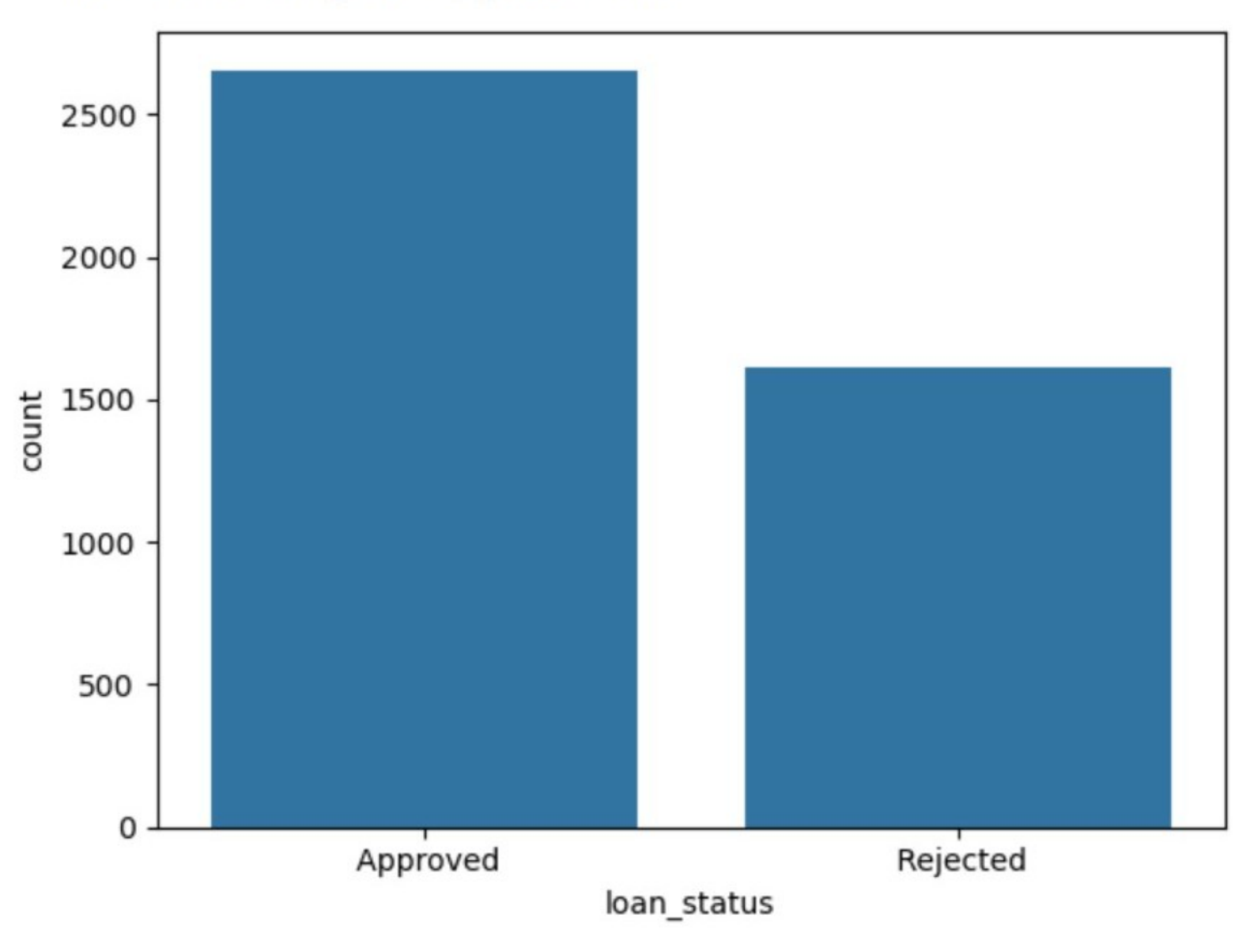
```python
#df.columns=df.columns.str.strip().str.replace(" ","")
```

In [319…

```python
sb.countplot(data=df,x='loan_status')
```

Out[319…

```
<Axes: xlabel='loan_status', ylabel='count'>
```



In [321…

```python
df.columns
```

Out[321…

```
Index(['loan_id', 'no_of_dependents', 'education', 'self_employed',
       'income_annum', 'loan_amount', 'loan_term', 'cibil_score',
       'residential_assets_value', 'commercial_assets_value',
       'luxury_assets_value', 'bank_asset_value', 'loan_status'],
      dtype='object')
```

# CIBIL Score Vs Loan Status

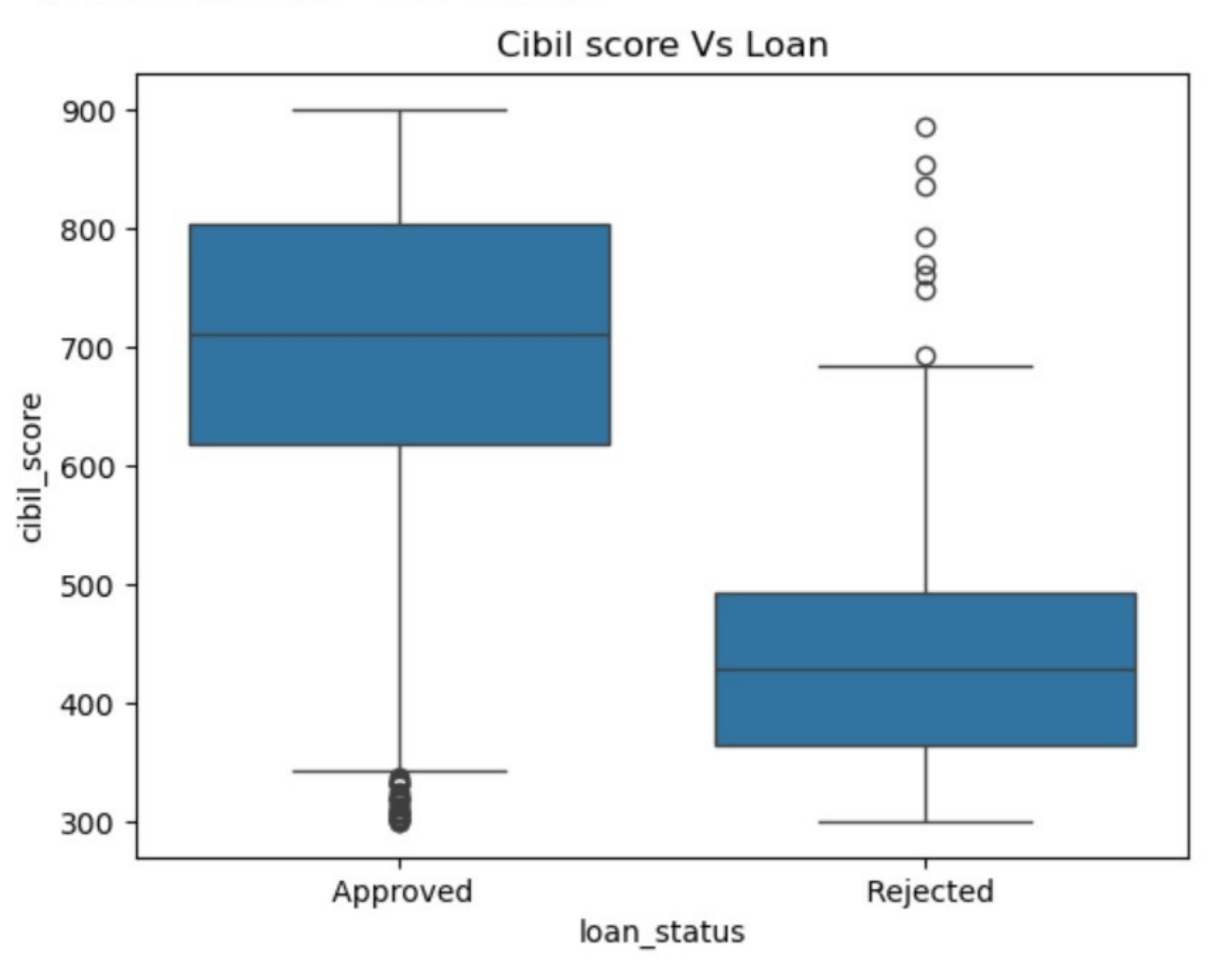
In [324…

```python
sb.boxplot(
    data=df,
    x='loan_status',
    y='cibil_score'
)

plt.title("Cibil score Vs Loan")
```

Out[324…

```
Text(0.5, 1.0, 'Cibil score Vs Loan')
```



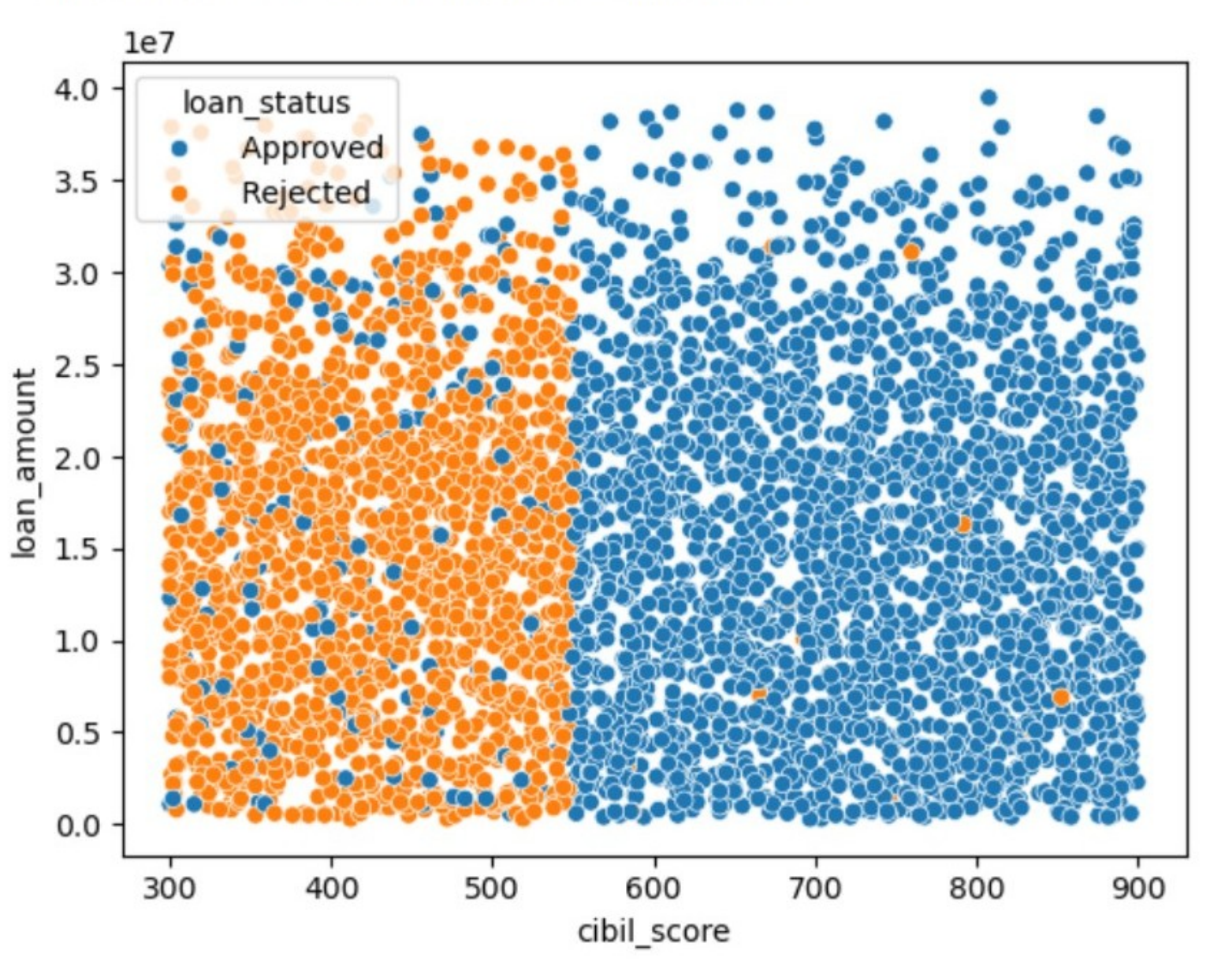
## Cibil score Vs Loan Amount

In [327…

```python
sb.scatterplot(
    data=df,
    x='cibil_score',
    y='loan_amount',
    hue='loan_status'
)
```

Out[327…

```
<Axes: xlabel='cibil_score', ylabel='loan_amount'>
```



In [328…

```python
df
```

Out[328…

| | loan_id | no_of_dependents | education | self_employed | income_annum | loan_amount | loan_term | cibil_score | residential_assets_value | commercial_assets_value | luxury_assets_value | bank_asset_value | loan_stat |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **0** | 1 | 2 | Graduate | No | 9600000 | 29900000 | 12 | 778 | 2400000 | 17600000 | 22700000 | 8000000 | Approv |
| **1** | 2 | 0 | Not Graduate | Yes | 4100000 | 12200000 | 8 | 417 | 2700000 | 2200000 | 8800000 | 3300000 | Reject |
| **2** | 3 | 3 | Graduate | No | 9100000 | 29700000 | 20 | 506 | 7100000 | 4500000 | 33300000 | 12800000 | Reject |
| **3** | 4 | 3 | Graduate | No | 8200000 | 30700000 | 8 | 467 | 18200000 | 3300000 | 23300000 | 7900000 | Reject |
| **4** | 5 | 5 | Not Graduate | Yes | 9800000 | 24200000 | 20 | 382 | 12400000 | 8200000 | 29400000 | 5000000 | Reject |
| **...** | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | |
| **4264** | 4265 | 5 | Graduate | Yes | 1000000 | 2300000 | 12 | 317 | 2800000 | 500000 | 3300000 | 800000 | Reject |
| **4265** | 4266 | 0 | Not Graduate | Yes | 3300000 | 11300000 | 20 | 559 | 4200000 | 2900000 | 11000000 | 1900000 | Approv |
| **4266** | 4267 | 2 | Not Graduate | No | 6500000 | 23900000 | 18 | 457 | 1200000 | 12400000 | 18100000 | 7300000 | Reject |
| **4267** | 4268 | 1 | Not Graduate | No | 4100000 | 12800000 | 8 | 780 | 8200000 | 700000 | 14100000 | 5800000 | Approv |
| **4268** | 4269 | 1 | Graduate | No | 9200000 | 29700000 | 10 | 607 | 17800000 | 11800000 | 35700000 | 12000000 | Approv |

4269 rows × 13 columns

In [330…

```python
df=df.drop('loan_id',axis=1)
```

In [333…

```python
df
```

Out[333…

| | no_of_dependents | education | self_employed | income_annum | loan_amount | loan_term | cibil_score | residential_assets_value | commercial_assets_value | luxury_assets_value | bank_asset_value | loan_status |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **0** | 2 | Graduate | No | 9600000 | 29900000 | 12 | 778 | 2400000 | 17600000 | 22700000 | 8000000 | Approved |
| **1** | 0 | Not Graduate | Yes | 4100000 | 12200000 | 8 | 417 | 2700000 | 2200000 | 8800000 | 3300000 | Rejected |
| **2** | 3 | Graduate | No | 9100000 | 29700000 | 20 | 506 | 7100000 | 4500000 | 33300000 | 12800000 | Rejected |
| **3** | 3 | Graduate | No | 8200000 | 30700000 | 8 | 467 | 18200000 | 3300000 | 23300000 | 7900000 | Rejected |
| **4** | 5 | Not Graduate | Yes | 9800000 | 24200000 | 20 | 382 | 12400000 | 8200000 | 29400000 | 5000000 | Rejected |
| **...** | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... |
| **4264** | 5 | Graduate | Yes | 1000000 | 2300000 | 12 | 317 | 2800000 | 500000 | 3300000 | 800000 | Rejected |
| **4265** | 0 | Not Graduate | Yes | 3300000 | 11300000 | 20 | 559 | 4200000 | 2900000 | 11000000 | 1900000 | Approved |
| **4266** | 2 | Not Graduate | No | 6500000 | 23900000 | 18 | 457 | 1200000 | 12400000 | 18100000 | 7300000 | Rejected |
| **4267** | 1 | Not Graduate | No | 4100000 | 12800000 | 8 | 780 | 8200000 | 700000 | 14100000 | 5800000 | Approved |
| **4268** | 1 | Graduate | No | 9200000 | 29700000 | 10 | 607 | 17800000 | 11800000 | 35700000 | 12000000 | Approved |

4269 rows × 12 columns

In [335…

```python
df.columns=df.columns.str.strip()
```

In [337…

```python
df['education'].unique()
```

Out[337…

```
array([' Graduate', ' Not Graduate'], dtype=object)
```

In [339…

```python
df['education']=df['education'].map({
    ' Graduate':1,
    ' Not Graduate':0
})
```

In [341…

```python
df
```

Out[341…

| | no_of_dependents | education | self_employed | income_annum | loan_amount | loan_term | cibil_score | residential_assets_value | commercial_assets_value | luxury_assets_value | bank_asset_value | loan_status |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **0** | 2 | 1 | No | 9600000 | 29900000 | 12 | 778 | 2400000 | 17600000 | 22700000 | 8000000 | Approved |
| **1** | 0 | 0 | Yes | 4100000 | 12200000 | 8 | 417 | 2700000 | 2200000 | 8800000 | 3300000 | Rejected |
| **2** | 3 | 1 | No | 9100000 | 29700000 | 20 | 506 | 7100000 | 4500000 | 33300000 | 12800000 | Rejected |
| **3** | 3 | 1 | No | 8200000 | 30700000 | 8 | 467 | 18200000 | 3300000 | 23300000 | 7900000 | Rejected |
| **4** | 5 | 0 | Yes | 9800000 | 24200000 | 20 | 382 | 12400000 | 8200000 | 29400000 | 5000000 | Rejected |
| **...** | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... |
| **4264** | 5 | 1 | Yes | 1000000 | 2300000 | 12 | 317 | 2800000 | 500000 | 3300000 | 800000 | Rejected |
| **4265** | 0 | 0 | Yes | 3300000 | 11300000 | 20 | 559 | 4200000 | 2900000 | 11000000 | 1900000 | Approved |
| **4266** | 2 | 0 | No | 6500000 | 23900000 | 18 | 457 | 1200000 | 12400000 | 18100000 | 7300000 | Rejected |
| **4267** | 1 | 0 | No | 4100000 | 12800000 | 8 | 780 | 8200000 | 700000 | 14100000 | 5800000 | Approved |
| **4268** | 1 | 1 | No | 9200000 | 29700000 | 10 | 607 | 17800000 | 11800000 | 35700000 | 12000000 | Approved |

4269 rows × 12 columns

In [343…

```python
df['self_employed'].unique()
```

Out[343…

```
array([' No', ' Yes'], dtype=object)
```

In [345…

```python
df['self_employed']=df['self_employed'].map({
    ' Yes':1,
    ' No':0
})
```

In [347…

```python
df
```

Out[347…

| | no_of_dependents | education | self_employed | income_annum | loan_amount | loan_term | cibil_score | residential_assets_value | commercial_assets_value | luxury_assets_value | bank_asset_value | loan_status |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **0** | 2 | 1 | 0 | 9600000 | 29900000 | 12 | 778 | 2400000 | 17600000 | 22700000 | 8000000 | Approved |
| **1** | 0 | 0 | 1 | 4100000 | 12200000 | 8 | 417 | 2700000 | 2200000 | 8800000 | 3300000 | Rejected |
| **2** | 3 | 1 | 0 | 9100000 | 29700000 | 20 | 506 | 7100000 | 4500000 | 33300000 | 12800000 | Rejected |
| **3** | 3 | 1 | 0 | 8200000 | 30700000 | 8 | 467 | 18200000 | 3300000 | 23300000 | 7900000 | Rejected |
| **4** | 5 | 0 | 1 | 9800000 | 24200000 | 20 | 382 | 12400000 | 8200000 | 29400000 | 5000000 | Rejected |
| **...** | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... |
| **4264** | 5 | 1 | 1 | 1000000 | 2300000 | 12 | 317 | 2800000 | 500000 | 3300000 | 800000 | Rejected |
| **4265** | 0 | 0 | 1 | 3300000 | 11300000 | 20 | 559 | 4200000 | 2900000 | 11000000 | 1900000 | Approved |
| **4266** | 2 | 0 | 0 | 6500000 | 23900000 | 18 | 457 | 1200000 | 12400000 | 18100000 | 7300000 | Rejected |
| **4267** | 1 | 0 | 0 | 4100000 | 12800000 | 8 | 780 | 8200000 | 700000 | 14100000 | 5800000 | Approved |
| **4268** | 1 | 1 | 0 | 9200000 | 29700000 | 10 | 607 | 17800000 | 11800000 | 35700000 | 12000000 | Approved |

4269 rows × 12 columns

In [349…

```python
df['loan_status'].unique()
```

Out[349…

```
array([' Approved', ' Rejected'], dtype=object)
```

In [351…

```python
df['loan_status']=df['loan_status'].map({
    ' Approved':1,
    ' Rejected':0
})
```

In [353…

```python
df
```